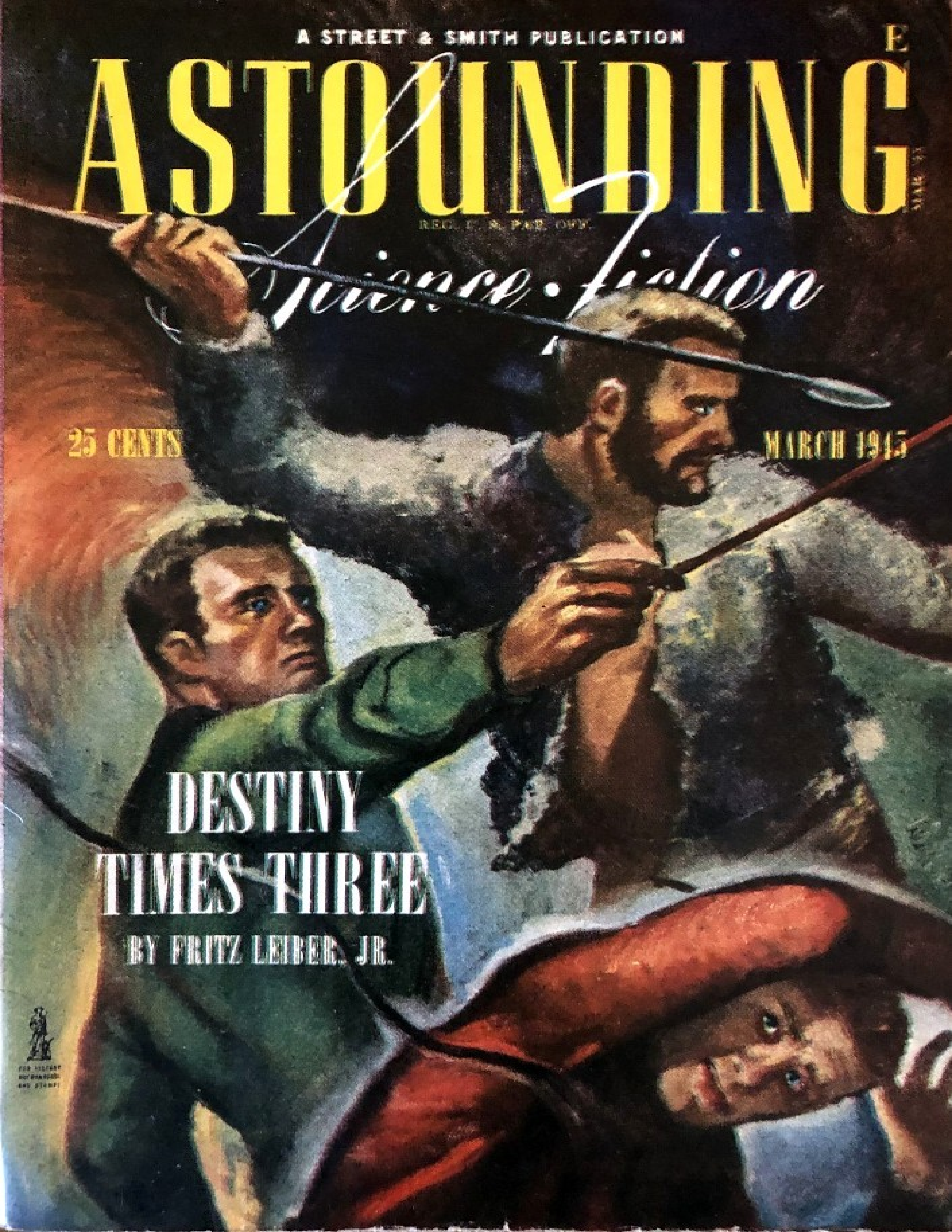
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Special Delivery

By GEORGE O. SMITH

Illustrated by Kramer

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Don Channing grinned at his wife knowingly. Arden caught his glance and then laughed. Walt Franks leaned back and looked highly superior. "Go ahead and laugh, darn you. I tell you it can be done."

"Walt, ever since you tried that stunt of aerating soap with hydrogen to make a floating soap for shower baths, I've been wondering about your kind of genius."

"Oh no," objected Arden.

"Well, he wondered about it after nearly breaking his neck one morning."

"That I did," grinned Walt. "It's still a good idea."

"But the idea of transmitting matter is fantastic."

"Agreed," admitted Walt. "But so is the idea of transmitting power."

"It would come in handy if possible," remarked Don. "At slightly under 2-G, it takes only four hours to make Luna from Terra. On the other hand, shipping stuff from Melbourne, Australia, to New York City, or to the Mojave Spaceport takes considerably longer. With spacecraft as super stratosphere carriers it isn't too good, because you've got to run in a circle. In space you run at constant acceleration to mid-point and then decelerate the rest of the way. Fine for mile-eating, but not too hot for cutting circles."

"Well, having established the need of a matter transmitter, now what?"

"Go on, Walt. You're telling us."

"Well," said Walt, penciling some notes on the tablecloth, "it's like this. The Carroll-Baler power-transmission tube will carry energy. According to their initial experiments, they had some trouble."

"They had one large amount, if I recall."

"Specifically, I recall the incident of the hammer. Remember?"

"Barney Carroll got mad and swung a hammer at the tube, didn't he?"

"It was one of them, I don't recall which."

"No matter of importance," said Don. "I think I know what you mean. He hit the intake end—or tried to. The hammer was cut neatly and precisely off, and the energy of the blow was transmitted, somehow, to the wall."

"Through the wall," corrected Walt. "It cracked the plaster, but it went through so fast that it merely cracked it. The main blow succeeded in breaking the marble facade of the city hall."

"Um. Now bring us up to date. What have you in mind?"

"A tube which scans matter atom by atom, line by line, and plane by plane. The matter is removed, atom by atom, and transmitted to a sort of matter bank in the instrument."

"A what?"

"Matter bank," said Walt. "We can't transmit the stuff itself. That's out. We can't dissipate the atomic energy or whatever effect we might get. We can establish a balance locally by using the energy release to drive the restorer. According to some initial experiments, it can be done. We take something fairly complex and break it down. We use the energy of destruction to re-create the matter in a bank, or solid block of local stuff. Let it be a mass of stuff if it wants to, at any rate, the signal impulses from the breakdown will be transmitted, scanned, if you will, and transmitted to a receiver which reverses the process. It scans, and the matter bank is broken down and the object is rebuilt.

"I hope we can get free and unrestricted transmutation," offered Don. "You can't send a steel spring out and get one back made of copper."

"I get your point."

"The space lines will hate you," said Arden.

"Too bad. I wonder if it'll carry people."

"Darling," drawled Arden, "don't you think you'd better catch your rabbit first?"

"Not too bad a thought," agreed Don. "Walt, have you got any rabbit traps out?"

"A couple. I've been tinkering a bit. I know we can disintegrate matter through a power tube of slight modification, and reintegrate it with another. At the present state of the art, it is a mess."

"A nice mess," laughed Don. "Go ahead, though. We'll pitch in when the going gets hard."

"That's where I stand now. The going is tough."

"What's the trouble?"

"Getting a perfect focus. I want it good enough so that we can scan a polished sheet of steel—and it'll come out as slick as the original."

"Naturally. We'd better get Wes Farrell on the job."

"I wonder what by-product we'll get this time."

"Look, Walt. Quit hoping. If you get this thing running right, it'll put your name in history."

"After all," grinned Walt, "I've got to do something good enough to make up for that Channing Layer."

"Kingman is still fuming over the Channing Layer. Sometimes I feel sorry that I did it to him like that."

"Wasn't your fault, Don. You didn't hand him the thing knowing that the Channing Layer would inhibit the transmission of energy. It happened. We get power out of Sol—why shouldn't they? They would, except for the Channing Layer."

"Wonder what your idea will do."

"About the Channing Layer? Maybe your space-line competition is not as good as it sounds."

"Well, they use the power-transmission tubes all over the face of the Solar System. I can't see any reason why they couldn't ship stuff from Sidney to Mojave and then space it out from there."

"What an itinerary! By Franks' matter transmitter to Mojave. Spacecraft to Luna. More matter transmission from Luna to Phobos. Then transshipped down to Lincoln Head, and by matter transmitter to Canalopsis. *Whoosh!*"

"Do we have time to go into the old yarn about the guy who listened in and got replicas?" asked Arden.

"That's a woman's mind for you," grinned Channing. "Always making things complicated. Arden, my lovely but devious-minded woman, let's wait until we have the spry beastie by the ears before we start to make rabbit pie."

"It's not as simple as it sounds," warned Walt. "But it's there to worry about."

"But later. I doubt that we can reason that angle out."

"I can," said Arden. "Can we tap the power beams?"

"Wonderful is the mind of woman!" praised Don. "Positively wonderful! Arden, you have earned your next fur coat. Here I've been thinking of radio transmission all the time. No, Arden, when you're set up for sheer energy transmission, it's strictly no dice. The crimped-up jobs we use for communications can be tapped—but not the power-transmission beams. If you can keep the gadget working on that line, Walt, we're in and solid."

"I predict there'll be a battle. Are we shipping energy or communications?"

"Let Kingman try and find a precedent for that. Brother Blackstone himself would be stumped to make a ruling. We'll have to go to work with the evidence as soon as we get a glimmer of the possibilities. But I think we have a good chance. We can diddle up the focus, I'm certain."

Arden glowered. "Go ahead—have your fun. I see another couple of weeks of being a gadgeteer's widow." She looked at Walt Franks. "I could stand it if the big lug only didn't call every tool, every part, and every effect either *she* or *baby*!"

Walt grinned. "I'd try to keep you from being lonely, but I'm in this too, and besides, you're my friend's best wife."

"Shall we drag that around a bit? I think we could kill a couple of hours with it sometime."

"Let it lie there and rot," snorted Channing cheerfully. "We'll pick it up later. Come on, Walt. We've got work to do."

Mark Kingman glowered at the 'gram and swore under his breath. He wondered whether he might be developing a persecution complex; it seemed as though every time he turned around, Venus Equilateral was in his hair, asking for something or other. And he was not in any position to quibble about it. Kingman was smart enough to carry his tray very level. Knowing that they were waiting for a chance to prove that he had been connected with the late Hellion Murdoch made him very cautious. There was no doubt in any mind that Murdoch was written off the books. But whether Murdoch had made a sufficiently large impression on the books of Terran Electric to have the connection become evident—that worried Kingman.

So he swore at each telegram that came in, and then sent the desired object out with the next ship. Compared to his former attitude toward Venus Equilateral, Mark Kingman was behaving like an honor student in a Sunday school.

Furthermore, behaving himself did not make him feel good.

He punched the buzzer, told his secretary to call in the shop foreman, and then sat back and wondered about the 'gram.

He was still wondering when the man entered. Kingman looked up and fixed his superintendent with a fish glance. "Horman, can you guess why the Venus Equilateral crowd would want two dozen gauge blocks?"

"Sure. We use Johannson Blocks all the time."

"Channing wants twenty-four blocks. All three inches on a side—cubes. Square to within thirty seconds of angle, and each of the six faces optically flat to one quarter wave length of Cadmium light."

"*Whoosh!*" said Horman. "I presume the three-inch dimension must be within a half wave length?"

"They're quite lenient," said Kingman bitterly. "A full wave length!"

"White of them," grunted Horman. "I suppose the same thing applies?"

"We're running over thin ice," said Kingman reflectively. "I can't afford to play rough. We'll make up their blocks."

"I wonder what they want 'em for."

"Something tricky, I'll bet."

"But what could you use two dozen gauge blocks for? All the same size."

"Inspection standards?" asked Kingman.

"Not unless they're just being difficult. You don't put primary gauges on any production line. You make secondary gauges for production line use and keep a couple of primaries in the check room to try the secondaries on. In fact, you usually have a whole set of gauge blocks to build up to any desired dimension so that you don't have to stock a half-million of different sizes."

"It's possible that they may be doing something extremely delicate?"

"Possible," said Horman slowly. "But not too probable. On the other hand, I may be one hundred percent wrong. I don't know all the different stuff a man can make, by far. My own experience indicates that nothing like that would be needed. But that's just one man's experience."

"Channing and that gang of roughneck scientists have been known to make some fancy gadgets," said Kingman grudgingly.

"If you'll pardon my mentioning the subject," said Horman in a scathing tone, "you'd have been far better off to tag along with 'em instead of fighting 'em."

"I'll get 'em yet!"

"What's it got you so far?"

"I'm not too bad off. I've come up from the assistant chief legal counsel of Terran Electric to controlling the company."

"And Terran Electric has slid down from the topmost outfit in the system to a seventh rater."

"We'll climb back. At any rate, I'm better off personally. You're better off personally. In fact, everybody that had enough guts to stay with us is better off."

"Yeah—I know. It sounds good on paper. But make a bum move again, Kingman, and we'll all be in jail. You'd better forget that hatred against Venus Equilateral and come down to earth."

"Well, I've been a good boy for them once. After all, I did point out the error in their patent on the solar beam."

"That isn't all. Don't forget that Terran Electric's patent was at error too."

"Frankly it was a minor error. It's one of those things that is easy to get caught on. You know how it came about?"

"Nope. I accepted it just like everybody else. It took some outsider to laugh at me and tell me why."

Kingman smiled. "It's easy to get into easy thinking. They took power from Sirius—believe it or not—and then made some there-and-back time measurements and came up with a figure that was about the square of one hundred eighty-six thousand miles per second. But you know that you can't square a velocity and come up with anything that looks sensible. The square of a velocity must be some concept like an expanding area."

"Or would it be two spots diverging along the sides of a right angle?" queried Horman idly. "What was their final answer?"

"The velocity of light is a concept. It is based on the flexibility of space—its physical constants, so to speak. Channing claims that the sub-etheric radiation bands of what we have learned to call the driver radiation propagates along some other medium than space itself. I think they were trying to establish some mathematical relation—which might be all right, but you can't establish that kind of relation and hope to hold it. The square of C in meters comes out differently than the square of C in miles, inches, or a little-used standard, the light-second, in which the velocity of light is unity, or One. Follow? Anyway, they made modulation equipment of some sort and measured the velocity and came up with a finite figure which is slightly less than the square of one hundred eighty-six thousand miles per second. Their original idea was wrong. It was just coincidence that the two figures came out that way. Anyway," smiled Kingman, "I pointed it out to them and they quick changed their patent letters. So, you see, I've been of some help."

"Nice going. Well, I'm going to make those gauges. It'll take us one long time, too. Johannson Blocks aren't the easiest thing in the world to make."

"What would you make secondary standards out of?"

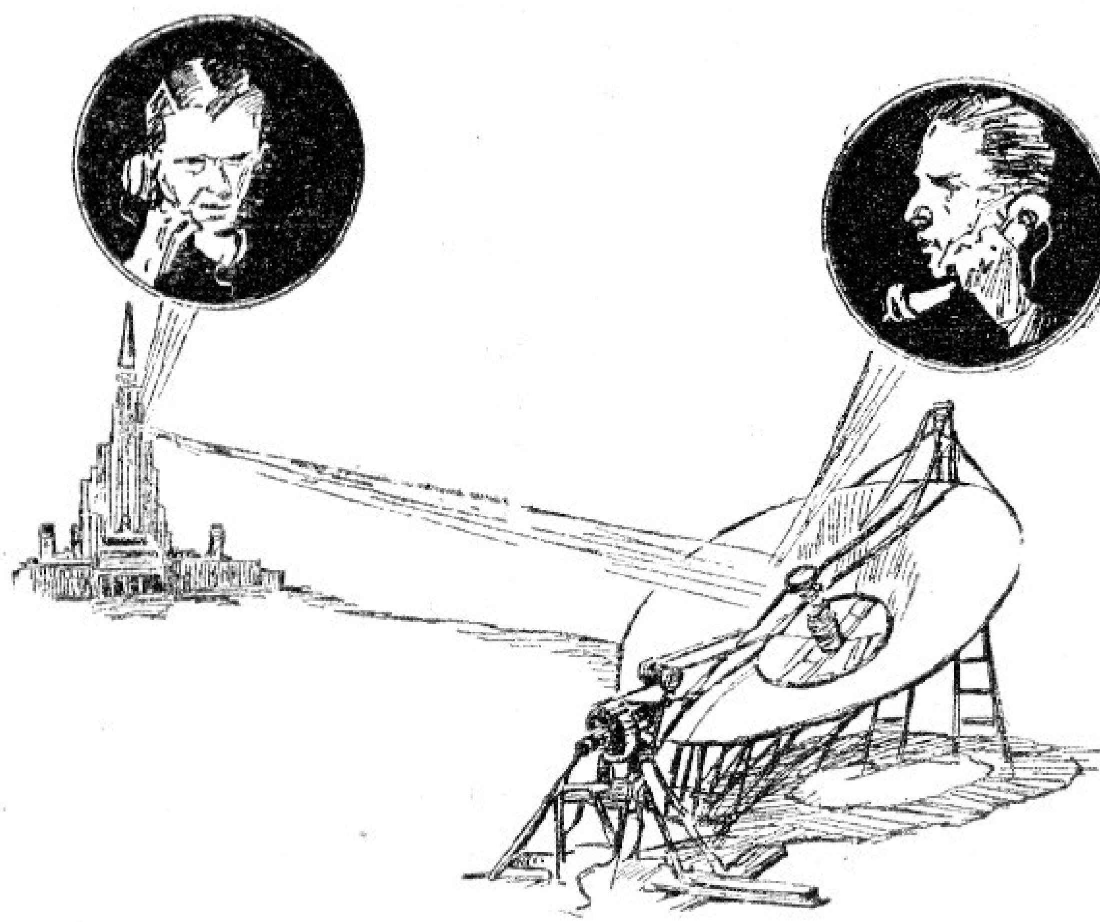
"We use glass gauges, mostly. They don't dinge or bend when dropped—they go to pieces or not at all. We can't have a bent gauge rejecting production parts, you know, and steel gauges can be bent. Besides, you can grind glass to a half wave length of light with ease, but polishing steel is another item entirely."

"I'm going to call Channing and ask him about glass blocks. It may be that he might use them. Plus the fact that I may get an inkling of the ultimate use. They have no production lines running on Venus Equilateral, have they?"

"Nope. Not at all. They're not a manufacturing company."

"Well, I'm going to call."

Kingman's voice raced across Terra to Hawaii, went on the communications beams of the sky-pointing reflectors, and rammed through the Heaviside Layer to Luna. At the Lunar Station, his voice was mingled in multiplex with a thousand others and placed on the sub-ether beams to Venus Equilateral.



Don Channing answered the 'phone. "Yes?"

"Kingman, Dr. Channing."

Don grunted. He did not care to be addressed by title when someone who disliked him did it. His friends did not, and Kingman's use of the title made it an insult.

"Look," said Kingman, "what do you want to use those blocks for?"

"We've got a job of checking dimensions."

"Nothing more? Do you need the metal for electrical reasons?"

"No," said Don. "What have you in mind?"

"Our toolshop is nicely equipped to grind glass gauges. We can do that better than making Jo-blocks. Can you use glass ones?"

"Hang on a minute." Channing turned to Walt. "Kingman says his outfit uses glass gauges. Any reason why we can't?"

"See no reason why not. I've heard of using glass gauges, and they've got some good reasons, too. Tell him to go ahead."

"Kingman? How soon can we get glass ones?"

"Horman, how soon on the glass blocks?"

"Two dozen? About a week."

"We'll have your blocks on the way within four days, Channing. Four days minimum, plus whatever wait is necessary to get 'em aboard a spacer."

"We'll check from this end on schedules. We need the blocks, and if the wait is too long, we'll send the *Relay Girl* in for 'em."

Don hung up and then said: "Glass ones might be a good idea. We can check the transmission characteristics optically. I think we can check more, quickly, than by running analysis on steel."

"Plus the fact that you can get the blocks back after test," grinned Walt. "Once you tear into a steel block to check its insides, you've lost your sample. I don't know any other way to check the homogeneity than by optical tests."

"O.K. Well, four days for glass will do better than a couple of months on steel blocks."

"Right. Now let's look up Wes and see what he's come up with."

They found Farrell in one of the blister laboratories, working on a small edition of the power-transmission tubes. He was not dressed in spacesuit, and so they entered the blister and watched him work.

"Have a little trouble getting the focus to stay sharp through the trace," complained Wes. "I can get focus of atomic proportions—the circle of confusion is about the size of the atom nucleus, I mean—at the axis of the tube. But the deflection of the cone of energy produces aberration, which causes coma at the edges. The corners of an area look fierce."

"I wonder if mechanical scanning wouldn't work better."

"Undoubtedly. You don't hope to send life, do you?"

"It would be nice—but no more fantastic than this thing is now. What's your opinion?"

Wes loosened a set screw on the main tube anode and set the anode forward a barely perceptible distance. He checked it with a vernier rule and tightened the screw. He made other adjustments on the works of the tube itself, and then motioned outside. They left the blister, Wes closed the airtight, and cracked the valve that let the air out of the blister. He snapped the switch on the outside panel and then leaned back in his chair while the cathode heated.

"With electrical scanning, you'll have curvature of field with this gadget. That isn't too bad, I suppose, because the restorer will have the same curvature. But you're going to scan three ways, which means correction for the linear distance from the tube as well as the other side deflections and their aberrations. Now if we could scan the gadget mechanically, we'd have absolute flatness of field, perfect focus, and so forth."

Walt grinned. "Thinking of television again? Look, bright fellows, how do you move an assembly of mechanical parts in quanta of one atomic diameter? They've been looking for that kind of gadget for centuries. Dr. Rowland and his gratings would turn over in their graves with a contrivance that could rule lines one atom apart."

"On what?" asked Don.

"If it would rule one atom lines, brother, you could put a million lines per inch on anything rulable with perfection, ease, eclat, and savoir faire. You follow my argument? Or would you rather take up this slip of my tongue and make something out of it?"

"O.K., fella. I see your point. How about that one, Wes?"

Wes Farrell grinned. "Looks like I'll be getting perfect focus with the electrical system here. I hadn't considered the other angle at all, but it looks a lot tougher than I thought."

He squinted through a wall-mounted telescope at the set-up on the inside of the blister. "She's hot," he remarked quietly, and then set to checking the experiment. Fifteen minutes of checking, and making notes, and he turned to the others with a smile. "Not too bad that way," he said.

"What are you doing?"

"I've established a rather complex field. In order to correct the aberrations, I've got nonlinear focusing fields in the places where they tend to correct for the off-axis aberration. To correct for the height-effect, I'm putting a variable corrector to control the whole cone of energy, stretching it or shortening it according to the needs. I think if I use a longer focal length I'll be able to get the thing running right.

"That'll lessen the need for correction, too," he added, cracking the blister-intake valve and letting the air hiss into the blister. He opened the door and went inside, and began to adjust the electrodes. "You know," he added over his shoulder, "we've got something here that might bring in a few dollars on the side. This matter-bank affair produces clean, clear, and practically pure metal. You might be able to sell some metal that was rated 'pure' and mean it."

"You mean absolutely, positively, guaranteed, uncontaminated, unadulterated, perfectly chemically pure?" grinned Don.

"Compared to what 'Chemically Pure' really means, your selection of adjectives is a masterpiece of understatement," laughed Walt.

"I'm about to make one more try," announced Wes. "Then I'm going to drop this for the time being. I've got to get up to the machine shop and see what they're doing with the rest of the thing."

"We'll take over that if you wish," said Don.

"Will you? I'll appreciate it. I sort of hate to let this thing go when I feel that I'm near an answer."

"We'll do it," said Walt. "Definitely."

They left the laboratory and made their way to the elevator that would lift them high into the Relay Station where the machine shop was located. As they entered the elevator, Don shook his head.

"What's the matter?"

"Well, Friend Farrell is on the beam again. If he feels that we're close to the answer, I'll bet a hat that we're hanging right on the edge. Also, that kind of work would kill me dead. He likes to stick on one thing until the bitter end, no matter how long it takes. I couldn't do it."

"I know. About three hours of this and you're wanting another job to clear your mind. Then you could tackle that one for about three hours and take back on the first."

"Trying to do that to Farrell would kill both him and the jobs," said Don. "But you and I can keep two or three projects going strong. Oh well, Wes is worth a million."

"He's the best we've got," agreed Walt. "Just because he has a peculiar slant on life is no sign he's not brilliant."

"It's you and I that have the cockeyed slant on life," grinned Don. "And frankly, I'm proud of it." He swung the elevator door aside and they walked down the corridor. "This isn't going to be much to see, but we'll take a look."

The machine shop, to the man, was clustered around the one cabinet under construction. They moved aside to permit the entry of Channing and Franks.

"Hm-m-m," said Don. "Looks like a refrigerator and incinerator combined."

It did. It stood five feet tall, three feet square, and was sealed in front by a heavy door. There was a place intended for the tube that Farrell was tinkering with in the blister, and the lines to supply the power were coiled behind the cabinet.

"Partly wired?" asked Don.

"Just the power circuits," answered Walton. "We'll have this finished in a couple of days more. The other one is completed except for Wes Farrell's section."

Channing nodded, and said: "Keep it going." He turned to Walt and after the passage of a knowing glance, the pair left. "Walt, this waiting is getting on my nerves. I want to go down to Joe's and drink myself into a stupor which will last until they get something cogent to work on."

"I'm with you, but what will Arden say?"

"I'm going to get Arden. Self-protection. She'd cut my feet off at the knees if I went off on a tear without her."

"I have gathered that," grinned Walt. "You're afraid of her."

"Yeah," drawled Don. "After all—she's the cook."

"I'm waiting."

"Waiting for what?"

"If and when. If you two go on as you have for another year without one of you turning up with a black eye, I may be tempted to go forth and track me down a dame of my own."

The cabinet stood in the north end of Venus Equilateral but it was not alone. It may even be the record for all times; certainly no other cabinet three by three by five ever had twenty-seven men all standing in a circle awaiting developments. The cabinet at the south end of Venus Equilateral was no less popular, though the number of watchers was less by one. Here, then, were winner and runner-up of inanimate popularity for the ages. The communicator system set in the walls of the two rooms carried sounds from the north room to the south, and those sounds in the south room could be heard in the north room. Channing grinned boyishly at Arden.

"This, my love, is a device which may make it quite possible for me to send you back to mother."

Arden smiled serenely. "No dice," she said, "Mother went back to grandmother last week. When is this thing going to cook?"

"Directly."

"What are we waiting for?"

"Walt."

"I'm ready," came Walt's voice through the speaker.

"About time, slowpoke."

"Really, it was not his fault," objected Wes. "I wanted to check the scanner-synchronization."

"He's precious," chortled Arden in Don's ear. "He wouldn't think of letting Walt—the big bum—take the blame for anything that wasn't Walt's fault."

"That's a good line," grinned Don. "Walt's faults. After we set this thing aside as a finished project, we'll set that 'Walt's faults' to music. Ready, Walt?"

"Right. I am now slipping the block into the cabinet. The door is closed. Have you got the preliminary synchronizing signal in tick?"

Channing called: "Wait a minute, I'm lagging a whole cycle."

"Cut your synchronization input and let the thing catch up."

"O. K. Um-m-m—Now, Walt."

"Has anyone any last words to say?" asked Walt.

No answer.

"Then since no one has any objections at this time, I assume that everything may be run off. Silence, people, we are going on the air!"

"There was a very faint odor of corn in Walt's last remark," said Don.

"I think the corn was on his breath," said Arden.

"Done!" announced Walt. "Don, crack the door so that the rest of us can laugh if it don't work."

Channing swaggered over and opened the door. He reached inside and took out the—object.

He held it up.

"Walt," said he, "what are you giving me?"

"Huh?"

"I presume that you shipped me one of the cubes?"

"Right."

"Well, what we got at this end would positively scare the right arm off of a surrealist sculptor."

"Hang on to it—I'll be up."

"Hang on to it?" laughed Don. "I'm afraid to."

It was three miles from one end of Venus Equilateral to the other and Walt made it in six minutes from the time he stepped into the little runaway car to the time he came into the north-end laboratory and looked over Channing's shoulder at the—thing—that stood on the table.

"Um," he said. "Sort of distorted, isn't it?"

"Quite," said Don. "This is glass. It was once a three-inch cube of precision, polish, and beauty. It is now a combination of a circular stairway with round corners and a sort of accordion pleat. Hell's bells!"

"Be not discouraged," gurgled Walt. "No matter what it looks like, we did transmit matter."

Arden tapped Don on the shoulder. "May I say it now?"

"You do—!"

"Then I won't say it doesn't matter."

"I'm ignoring your crude remark. Walt, we did accomplish something. It wasn't too good. Now let's figure out why this thing seems to have been run over with a fourth dimensional caterpillar-tread truck."

"Well, I can hazard a guess. The synchronizing circuits were not clamped perfectly. That gives the accordion-pleat effect. The starting of the trace was not made at the same place each time due to slippage. We'll have to beef up the synchronization impulse. The circular staircase effect was probably due to phase distortion."

"Could be," said Don. "That means we have to beef up the transmission band so it'll carry a higher frequency."

"A lower impedance with corrective elements?"

"Might work. Those will have to be matched closely. We're not transmitting on a line, you know. It's sheer transmission-tube stuff from here to there. Well, gang, we've had our fun. Now let's widen the transmission band and beef up the sync. Then we'll try number two."

Number two was tried the following afternoon. Again, everybody stood around and watched over Don's shoulder as he removed the cube from the cabinet.

"Nice," he said, doing a little war dance.

Franks came in puffing, took the cube from Don's fingers and inspected it. "Not too bad," he said.

"Perfect."

"Not by a jug full. The index of refraction is higher at this edge than at the other. See? Walt held the cube before a newspaper and they squinted through the glass block.

"Seems to be. Now why?"

"Second harmonic distortion, if present, would tend to thin out one side and thicken up the other side. A sine-wave transmission would result in even thicknesses, but if second harmonic distortion is present, the broad loops at the top create a condition where the average from zero to top is higher than the average from zero to the other peak. Follow?"

"That would indicate that the distortion was coming in at this end. If both were even, they would cancel."

"Right. Your scanning at one end is regular—at the other end it is irregular, resulting in non-homogeneity."

"The corners aren't really sharp," objected Arden.

"That's an easy one. The wave-front isn't sharp either. Instead of clipping sharply at the end of the trace, the signal tapers off. That means higher frequency response is needed."

"We need a term. Audio for sonics; radio for electronics, video for television signals—"

"Mateo," said Arden.

"Um—sounds sort of silly," grinned Walt.

"That's because it's strange. Mateo it is," said Don. "Our mateo amplifier needs higher frequency response in order to follow the square wave-front. Might put a clipper circuit in there, too."

"I think a clipper and sharpener will do more than the higher frequency," said Farrell. He was plying a vernier caliper, and he added: "I'm certain of that second harmonic stuff now. The dimension is cockeyed on this side. Tell you what, Don. I'm going to have the index of refraction measured within an inch of its life. Then we'll check the thing and apply some high-powered math and see if we can come up with the percentage of distortion."

"Go ahead. Meanwhile, we'll apply the harmonic analyzer to this thing and see what we find. If we square up the edges and make her homogeneous, we'll be in business."

"The space lines will hate you to pieces," said Arden.

"Nope. I doubt that we could send anything very large. It might be more bother to run a huge job than the money it costs to send it by spacer. But we have a market for small stuff that is hard to handle in space because of its size."

"I see no reason why Keg Johnson wouldn't go for a hunk of it," offered Wes Farrell.

"I've mentioned it to Keg; the last time I was in Canalopsis," said Walt. "He wasn't too worried—providing he could buy a hunk."

"Interplanet is pretty progressive," mused Don. "There'll be no reason why we can't make some real handy loose change out of this. Well, let's try again tomorrow."

"O. K. Let's break this up. Will we need any more blocks from Terran Electric?"

It was less than a month later that a newspaper reporter caught the advanced patent notice and swallowed hard. He did a double take, shook his head, and then read the names on the patent application and decided that someone was not fooling. He took leave and made the run to Venus Equilateral to interview the officials. He returned not only with a story, but with a sample glass block that he had seen run through the machine.

The news pushed one hatchet murder, a bank robbery, a football upset, and three political harangues all the way back to page seven. In terms more glowing than scientifically accurate, the matter transmitter screamed in three-inch headlines, trailed down across the page in smaller type, and was embellished with pictures, diagrams, and a description of the apparatus. The latter had been furnished by Walt Franks, and had been rewritten by the reporter because Walt's description was too dry.

The following morning Venus Equilateral had nine rush telegrams. Three were from cranks who wanted to go to Sirius and set up a restorer there to take people; four were from superstitious nuts who called Channing's attention to the fact that he was overstepping the rights given to him by his Creator; one was from a gentleman who had a number of ideas, all of which were based on the idea of getting something for nothing, and none of which were legal; and the last one was a rather curt note from Terran Electric, pointing out that this device came under the realm of the power-transmission tube and its developments and that they wanted a legal discussion.

"Have they got a leg to stand on?" asked Walt.

"I doubt it."

"Then to the devil with them," snapped Walt. "We'll tell 'em to go jump in the lake."

"Nope. We're going in to Terra and slip them the slug. If we clip them now, they'll have nothing to go on. If we wait until they get started, they'll have a fighting chance. Besides, I think that all they want to do is to have the facts brought out. Are we or are we not under the terms of that contract?"

"Are we?"

"We're as safe as Sol. And I know it. That contract pertained to the use of the Solar beam only, plus certain other concessions pertaining to the use of the power-transmission tubes and other basic effects as utilized in communications."

"Why can't we tell 'em that?"

"It's got to be told in a court of law," said Don. "Kingman's mind runs to legal procedure like Blackstone."

"We'll take the gadgets?"

"Right. What are you using for power?"

"What other? Solar beams, of course. We don't bother about running stuff around any more. We plug it in the 115-volt line, it energizes the little fellows just long enough to make them self-sustaining from Sol. All the 115-volt line does is to act as a starting circuit."

"You and Farrell had better dream up a couple of power supplies, then. We can't use the Solar beam on Terra."

"I know. We're a little ahead of you on that. Wes and one of the Thomas boys cooked up a beam-transducer power supply that will get its juice from any standard 115-volt, sixty-cycle line socket. We've two of them—and they run the things easily."

"Good. I'll 'gram Terran Electric and let 'em know we're on our way for the legal tangle. You load up the *Relay Girl* and we'll be on our way. Stock up the usual supply of bars, blocks, gadgets, and traps. Might include a bar magnet. When we show that it is still magnetized, we'll gain a point for sure."

"If we take a magnet, we'd better take the fluxmeter to show that the magnetic field hasn't dropped."

"Right. Take anything you can think of for a good show. We can knock them dead!"

Mark Kingman put his assistant legal counsel on the witness stand. "You will state the intent of the contract signed between Terran Electric and Venus Equilateral."

"The contract holds the following intent: Use of the power-transmission tubes for communications purposes shall fall under the jurisdiction of Venus Equilateral. For power transmission, the tubes and associated equipment is under the control of Terran Electric. In the matter of the Solar beam tubes, the contract is as follows: Venus Equilateral holds the control of the Solar beam in space, on man-made bodies in space, and upon those natural bodies in space where Venus Equilateral requires the Solar power to maintain subsidiary relay stations."

"Please clarify the latter," said Kingman. "Unless it is your intent to imply that Terra, Mars, and Mercury fall under the classification of 'places where Venus Equilateral requires power.'"

"Their control on natural celestial objects extends only to their own installations and requirements. Basically, aside from their own power requirements, Venus Equilateral is not authorized to sell power. In short, the contract implies that the use of the sub-etheric phenomena is divided so that Venus Equilateral may use this region for communications, while Terran Electric uses the sub-ether for power. In space, however, Venus Equilateral holds the rights to the Solar beam."

Frank Tinkin, head legal man of Venus Equilateral, turned to Don and said: "We should have this in a technical court."

Don took his attention from the long discussion of the contract and asked: "Why not change?"

"Judges hate people who ask for change of court. It is bad for the requestee—and is only done when the judge is open to the question or disinterested—and also when the suspicion of dislike is less dangerous than the judge himself."

"Well, this should be in a technical court."

"Want to chance it?'

"I think so. This is more than likely to turn up with differential equations, physics experts, and perhaps a demonstration of atom-smashing."

Kingman finished his examination and turned away. The judge nodded sourly at Tinkin. "Cross-examination?"

Tinkin faced the witness, nodded, and then faced the court.

"The witness' statements regarding the contract are true. However, Judge Hamilton, I will attempt to show that this case is highly technical in nature and as such falls under the jurisdiction of the Technical Court. May I proceed?"

"Counsel for the plaintiff assures me that this is not truly a technical case," snapped Hamilton. "However if you can definitely prove that the case in point hinges on purely technical matters, what you say may be instrumental in having this hearing changed. Proceed."

"Thank you." Tinkin turned to the witness. "Exactly what is the point in question?"

"The point in question," said the witness, "is whether or not the matter transmitter falls under Terran Electric's contract or Venus Equilateral's contract."

"Isn't the question really a matter of whether the basic effect is technically communication or power transmission?"

"Objection!" barked Kingman. "The counsel is leading the witness."

"Objection permitted—strike the question from the record."

"I was merely trying to bring out the technical aspect of the case," explained Tinkin. "I'll rephrase the question. Is it not true that the contract between Terran Electric and Venus Equilateral is based upon a certain technology?"

"Certainly."

"Then if the case is based on technical aspects—?"

"Objection!" marked Kingman. "More than half of all manufacturing contracts are based upon technical background. I quote the case of Hines versus Ingall in which the subject matter was the development of a new type of calculating machine. This case was heard in a legal court and disposed of in the same."

"Objection permitted."

"No further examination," said Tinkin. He sat down and turned to Don. "We're in trouble, Hamilton does not like us."

"Well, we still have the whip hand."

"Right, but before we get done we'll have trouble with Hamilton."

"Before we get done, Kingman will have trouble with us," said Don.

Terran Electric's lawyer called Wes Farrell to the stand. "Mr. Farrell, you are employed by Venus Equilateral?"

"Yes."

"In what capacity?"

"As an experimental physicist."

"And as such, you were involved in some phases of the device under discussion?"

"I was," said Farrell.

"Does the device make use of the Solar beam?"

"It does but—"

"Thank you," interrupted Kingman.

"I'm not through," snapped Farrell. "The Solar beam is not integral."

"It is used, though."

"It may be removed. If necessary, we can have hand-generators supplied to generate the operating power."

"I see," said Kingman sourly. "The device itself is entirely new and basic?"

"Not entirely. The main components are developments of existing parts, specialized to fit the requirements."

"They are based on specifically what?"

"Certain effects noted in the power-transmission tubes plus certain effects noted in the Solar beam tubes."

"And which of these effects is more contributory?"

"Both are about equally responsible. One will be useless without the other."

Kingman turned to the judge. "I intend to show that the use of these effects is stated in the contract."

"Proceed."

"Was there any time during the development of the device any question of jurisdiction?"

"None whatever," said Farrell. "We knew how we stood."

"The statement is hearsay and prejudiced," stated Kingman.

"Strike it from the record," snapped Hamilton.

"It stands at 'none whatever,'" said Kingman.

The secretary nodded.

"Since absolutely no attention was paid to the terms of the contract, doesn't that imply that a certain ignorance of the terms might prevail?"

"Objection!" shouted Tinkin. "Counsel's question implies legal carelessness on the part of his opponent."

"How can you be aware of the ramifications of a contract that you do not read?" stormed Kingman.

"Objection overruled."

"May I take exception?" requested Tinkin.

"Exception noted. Counsel, will you rephrase your question so that no lack of foresight is implied?"

"Certainly," smiled Kingman. "How were you certain that you were within your rights?"

"If this plan had been open to any question, my superiors would not have permitted me—"

"That will not serve!" snapped Kingman. "You are making an implication—your testimony is biased."

"Naturally," barked Farrell. "No one but an idiot would claim to have no opinion."

"Does that include the court?" asked Kingman suavely.

"Naturally not," retorted Farrell. "I was speaking of interested parties."

"Let it pass. In other words, Dr. Farrell, you were never sure that you were within your rights?"

"I object!" exploded Tinkin. "Counsel is questioning a witness whose business is not legal matters on a subject which is legal in every phase."

"Objection sustained," said Hamilton wearily. The matter was dropped, but Kingman had gained his point. The item may never appear in the records, but it was present in the judge's mind.

"Dr. Farrell," said Kingman, "since you have no legal training, precisely what has been your education and background?"

"I hold a few degrees in physics, one in mathematics, and also in physical chemistry." Farrell turned to the judge. "Judge Hamilton, may I explain my position here?"

"You may."

"I have spent thirteen years studying physics and allied sciences. I believe that I stand fairly high among my fellows. Since no man may be capable in many arts, I believe that I have not been lax in not seeking degrees in law."

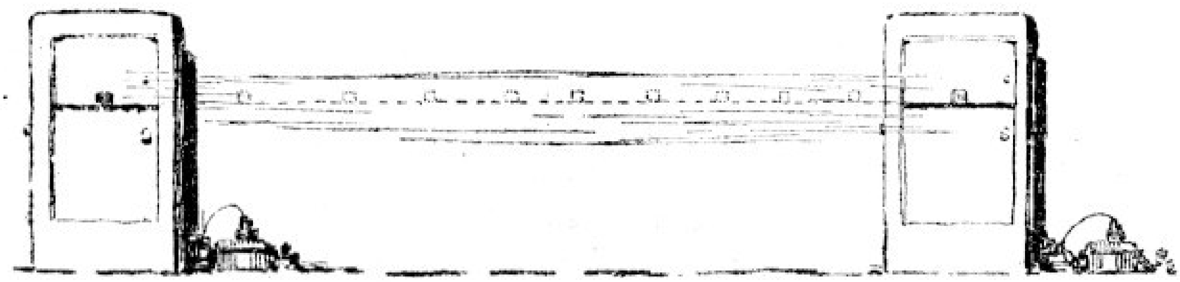
"No objection," said Kingman. "Dr. Farrell, in order that the process be properly outlined in the record, I am going to ask you to explain it in brief. How does your matter transmitter work?"

Farrell nodded, and took time to think. Tinkin whispered in Don's ear: "The stinker! He knows Hamilton hates anything more complex than a can opener!"

"What can we do?"

"Hope that our demonstration blasts them loose. That's our best bet, plus fighting for every inch."

Farrell moistened his lips and said: "Utilizing certain effects noted with earlier experimentation, we have achieved the following effects. The matter to be transmitted is placed in situ, where it is scanned by an atom-scanner. This tube removes the substance, atom by atom, converting the atoms to energy. This energy is then reconverted into atoms and stored in a matter bank as matter again. The energy of disintegration is utilized in reintegration at the matter bank with but small losses. Since some atoms have higher energy than others, the matter bank's composition will depend upon the scanned substance."



"The matter bank is composed of the same elements as the matter for transmission?" asked Kingman.

"No. Some elements release more energy than others. It is desirable that the energy-transfer be slightly negative. That is to say that additional energy must be used in order to make the thing work."

"Why?

"All power lines and other devices are developed for delivering energy, not receiving it. It is less disastrous to take energy from a power line than to try and drive it back in—and the energy must be dissipated somehow."

"Then the matter bank is not the same material."

"No," said Farrell. "The substance of the matter bank is nonhomogeneous. Instantaneously, it will be whatever element is necessary to maintain the fine balance of energy—and it is in constant change."

"Proceed," said Kingman.

"In passing from the disintegrator tube to reintegrator tube, the energy impresses its characteristic signal on a sub-ether transmission system. Radio might work, except that the signal is unbelievably complex. Wired communications—"

"Objection to the term," said Kingman.

"Sustained."

"Wired ... transfer? ... might work, but probably would not due to this same high complexity in transmitted signal. At any rate, upon reception, the signal is used to influence, or modulate, the energy passing from a disintegrator tube to a reintegrator tube in the receiver. But this time the tube is tearing down the matter bank and restoring the object. Follow?"

"I believe so. Does the Court understand?"

"This Court can follow the technical terms."

"Now, Dr. Farrell, the matter transmitter does actually transmit over a power-transmission tube?"

"Yes. Of the type developed by us for communications."

"But it is a power tube?"

"Yes."

"Then are you certain that you are sending no energy?"

"I object!" shouted Tinkin. "The question has no answer!"

"Hasn't it?" queried Kingman. "My worthy opponent, all questions have an answer."

"Objection overruled," snapped Hamilton sourly. "Let the witness answer."

"It is impossible to send communications without sending some energy. It is the intent to which the energy is put that determines the classification."

"Explain further."

"You must send energy when you communicate with a light-blinker," grinned Farrell, "The receiving party receives the energy, but couldn't possibly read a newspaper with it. The beams at Venus Equilateral send out several million watts—and by the time they get to Luna, they require amplifications bordering on the million-times before they are usable. The intent is clear—we are not supplying power, we are sending intelligence."

"I contend," said Kingman to the judge, "that the contract states clearly that developments of this device are to be used for communications only when operated by Venus Equilateral. I further contend that the transmission of matter does not constitute a communication, but rather a transfer of energy."

"I object," said Tinkin. "If this statement was objectionable to the learned counsel before, it is equally objectionable to me now."

"Previously," said Kingman suavely, "counsel was trying to influence a witness. I am merely trying to explain my point."

Hamilton cleared his throat. "Counsel is merely attempting to influence the Court; the same privilege will be available to his opponent at the proper time. That is why we have courts."

Tinkin sat down.

"I maintain that the concept of communication precludes matter transmission," stormed Kingman. "Matter transmission becomes a problem for the transportation companies and the power companies. Matter, your honor, is energy. They are transmitting energy!"

He stalked over to Tinkin and smiled affably. "Cross-examination?" he offered.

"No questions," said Tinkin.

Hamilton rapped on the bench. "Court is adjourned for ten minutes!"

"Looking for something?" asked Don. Arden turned from the window and faced him.

"I was trying to see Niagara Falls," she smiled. "I've heard that you could see 'em from Buffalo."

"You can," laughed her husband, "but not from this part of Buffalo. What do you want to see Niagara Falls for, anyway? Just a lot of water falling over a cliff at two pints to the quart."

"If you recall, chum, we went to Mars, not Niagara. There wasn't two pints of water on the whole planet, let alone a thing like Niagara."

Don nodded. "At the risk of offending a lot of Buffalonians, I'm beginning to dislike the place."

"It isn't the people," said Arden. "It's the position we're in. Bad, huh?"

"Not going too good at all. Kingman slips in a sly dig every now and then. Frankly, I am getting worried. He's got a few points that really hit very close to home. If he can sell the judge on a couple more of them, we'll be under the sod."

"You won't be out entirely, will you?"

"Not entirely. He'll have to use the beams of Venus Equilateral to operate, but he'll be collecting all the real gravy. We'll just be leasing our beams to him."

"Well, don't go down without a fight, chum."

"I won't. I really hate to see Kingman get ahead of this, though." Don stretched, took another look out across the city of Buffalo, and then said: "We'd best be getting back. We'll be late ... he said ten minutes."

They went down the staircase slowly, and at the courtroom door they met Keg Johnson. The latter smiled wearily. "Not too good?"

"Nope."

"Don, if you lose, then what?"

"Appeal, I guess."

"That isn't too good. Judges do not reverse lower courts unless a real miscarriage of justice takes place."

"I know, but that's our only chance."

"What would you advise me to do?"

"Meaning?" asked Don.

"Interplanet. We'll be run right out of business if this thing goes over to Kingman and that bunch."

"I know."

"Look, Don, have you tried living matter?"

"Plants go through with no ill effects. Microscopic life does, too. Animals we have tried usually die because of internal disorders—but they move while being scanned, and their bodies come out looking rather ugly. An anaesthetized mouse went through all right—lived for several hours. Died because the breathing-function made a microscopic rift in the lungs, and the beating heart didn't quite meet true. We must speed up the scanning-time to a matter of micro-seconds and then we can send living bodies with no harm."

"That would clean out the space lines," said Keg. "I think I'll offer that bird a slice of Interplanet for an interest, if he wins. We've got to have it, Don."

"I know, Keg. No hard feelings."

"Of course," said Keg wistfully. "We'll be across a barrel if you win, too. But the barrel will be less painful with you holding the handles than if Terran Electric holds them. The same offer goes for you, too."

"O.K.," nodded Channing. He turned and entered the courtroom.

Tinkin called Don Channing to the stand as his first witness. Don explained the function of Venus Equilateral, the job of interplanetary communications, and their work along other lines of endeavor. Then Tinkin said to the judge:

"I have here a glass cube, three inches on a side. This cube was transmitted from Venus Equilateral to the Lunar Station. I offer it as Exhibit A. It was a test-sample, and as you see, it emerged from the test absolutely perfect."

The judge took the cube, examined it with some interest, and then set it down on the desk.

"Now," said Tinkin, "if you do not object, I should like to present a demonstration of the matter transmitter. May I?"

Hamilton brightened slightly. "Permission is granted."

"Thank you." Tinkin made motions and the technicians came in with the two cabinets.

"This isn't good," said Kingman's assistant to the lawyer. "The old goat looks interested."

"Don't worry," said Kingman. "This'll take a long time, and by the time they get done, Hamilton will be ready to throw them out. Besides, it will make a good arguing point for my final blast. And, brother, I've got a talking-point that will scream for itself."

"But suppose they convince—"

"Look," smiled Kingman, "this is really no argument as to whether matter or intelligence is carried. Believe me, that has nothing to do with it. I'm keeping this one under the wraps until shooting-time so they won't be able to get an argument against it. We're a cinch. That's why I kept it in a legal court instead of a technical court. The Techs would award it to Channing on a technical basis, but the legal boys have got to follow my argument."

"How about an appeal?"

"The record of this court is still a very heavy argument. Look, they're about to start."

The racket and hubbub died, and Tinkin faced the judge. "These are plainly labeled. They are matter transmitter and matter receiver. We have here a set of metal bars. They are made of copper, steel, aluminum, some complex alloys, and the brother to that glass cube you have before you. We will transmit this set of objects from here to there. Have you any suggestions?"

"A matter of control and identity. What have you for control?"

"Nothing that is outside of our hands," smiled Tinkin. "Would you care to send something of your own? Your gavel? Inkwell? Marked coin? Anything?"

"I'd offer my glasses except for the fact that I can not see without them," said Judge Hamilton.

"We wouldn't break them or damage them a bit."

"I know—that much faith I have—but I'd not see the experiment."

"A good point. Anything else?"

"My watch. It is unique enough for me." He handed over the watch, which was quite sizable.

Tinkin inspected the watch and smiled. "Very old, isn't it? A real collector's item, I daresay."

Hamilton beamed. "There are nine of them in the Solar System," he said. "And I know where the other eight are."

"O.K., we'll put it on the top. I'll have to stop it, because the movement of the balance wheel would cause a rift during transmission."

"How about the spring tension?"

"No need to worry about that. We've sent loaded springs before. Now, people, stand back and we'll go on the air."

Don Channing himself inspected the machinery to see that nothing was wrong. He nodded at Walt Franks at the receiver, and then started the initial operations. "We are synchronizing the two machines," he said. "Absolute synchronization is necessary. Ready, Walt?"

"Right!"

Channing pushed a button. There was a minute, whirring hum, a crackle of ozone, very faint, and the almost-imperceptible wave of heat from both machines. "Now," said Walt Franks, "we'll see."

He opened the cabinet and reached in with a flourish.

His face fell. It turned rosy. He opened his mouth to speak, but nothing but choking sounds came forth. He spluttered, took a deep breath, and then shook his head in slow negation. Slowly, like a boy coming in for a whipping, Walt took out the judge's watch. He handed it to Don.

Don, knowing from Walt's expression that something was very, very wrong, took the watch gingerly, but quickly. He hated to look and was burning with worried curiosity at the same time.

In all three dimensions, the watch had lost its shape. It was no longer a lenticular object, but had a very faint sine wave in its structure. The round case was distorted in this wave, and the face went through the same long swell and ebb as the case. The hands maintained their distance from this wavy face by conforming to the sine-wave contour of the watch. And Channing knew without opening the watch that the insides were all created on the sine-wave principle, too. The case wouldn't have opened, Don knew, because it was a screw-on case, and the threads were rippling up and down along with the case and cover. The knurled stem wouldn't have turned, and as Channing shook the watch gently, it gave forth with one—and only one—tick as the slack in the distorted balance wheel went out.

He faced the judge. "We seem—"

"You blasted fools and idiots!" roared the judge. "Nine of them—!"

He turned and stiffly went to his seat. Channing returned to the witness chair.

"How do you explain that?" roared Judge Hamilton.

"I can only think of one answer," offered Channing in a low voice. "We made the power supplies out of power and voltage transducers and filtered the output for sixty cycles. Buffalo is still using twenty-five-cycle current. Since the reactances of both capacity and inductance vary according to the —"

"Enough of this!" roared Hamilton. "I—No, I may not say it. I am on the bench and what I am thinking would bring impeachment. Proceed Attorney Kingman."

Kingman took the cue, and before anyone realized that it was still Tinkin's floor, he opened.

"Dr. Channing, you can send a gallon of gasoline through this, ah, so-called matter transmitter?"

"Naturally."

"Then, your honor, it is my contention that no matter what the means or the intent, this instrument utilizes the sub-etheric effects to transmit energy! It is seldom possible to transmit power over the same carriers that carry communications—only very specialized cases prevail, and they are converted to the job. But this thing is universal. Perhaps it does transmit intelligence. It will and can be used to transmit energy! Matter, your honor, is energy! That, even the learned opponent will admit. We have our own means of transmitting power—this is another—and no matter what is intended, power and energy will be transmitted over its instruments.

"Since this machine transmits energy, I ask that you rule that it fall under that classification. I rest my case."

Hamilton nodded grumly. Then he fixed Tinkin with an ice-cold stare. "Have you anything to offer that may possibly be of any interest to me?"

Tinkin shook his head. He was still stunned.

"I shall deliver my ruling in the morning. I am overwrought and must rest. Adjourned until tomorrow morning."

The only sounds in the room were the tinkle of glassware and the occasional moan of utter self-dislike. Channing sat with his glass in his hand and made faces as he lifted it. Franks matched his mood. Both of them were of the type that drinks only when feeling good because it made them feel better. When they drank while feeling low, it made them feel lower, and at the present time they were about as far down as they could get. They knew it; they took the liquor more as a local anaesthetic than anything else. Arden, whose disappointment was not quite as personal as theirs, was not following them drink for drink, but she knew how they felt and was busying herself with glass, ice, and bottle as they needed it.

It was hours since the final let-down in the court. They knew that they could appeal the case, and probably after a hard fight they would win. It might be a year or so before they did, and in the meantime they would lose the initial control over the matter transmitter. They both felt that having the initial introduction in their hands would mean less headache than having Terran Electric exploit the thing to the bitter end as quickly as possible.

The fact of sunrise—something they never saw on Venus Equilateral—did not interest them one bit. It grew light outside, and as the first glimmerings of sunrise came, a knock on their door came also.

"Mice," hissed Walt.

"S'nock on door."

"Mice knocking on door?"

"Naw."

"Mice gnawing on door?"

"It's Wes Farrell," announced Arden, opening the door.

"Let'm in. S'all right, Wes. Anyone c'n make mishtake."

"He's sober."

"Gettum drink," said Don. "Gettum drink—gettum drunk."

"Look, fellows, I'm sorry about that fool mistake. I've been working on the judge's ticker. I've fixed it."

"Fitched it?" asked Walt, opening his eyes wide.

"Close 'em—Y'll bleed t' death," gurgled Don.

Farrell dangled the judge's watch before them. It was perfect. It ticked, it ran, and though they couldn't possibly have seen the hands from a distance of more than nine inches, it was keeping perfect time.

Don shook his head, moaned at the results of the shaking, put both hands on his head to hold it down, and looked again. "How'ja do it?"

"Made a recording of the transmitted signal. Fixed the power-supply filters first. Then took the recording—"

"On whut?" spluttered Walt.

"On a disk like the alloy-tuners in the communications beams. Worked fine. Anyway, I recorded the signal, and then started to buck out the ripple by adding some out-of-phase hum to cancel the ripple."

"Shounds reas'n'ble."

"Worked. I had a couple of messes, though."

"Messessesesss?" hissed Walt, losing control over his tongue.

"Yes. Had a bit of trouble making the ripple match." Farrell pulled several watches from his pocket. "This one added ripple. It's quite cockeyed. This one had cross-ripple and it's really a mess. It sort of looks like you feel, Walt. I've got 'em with double ripples, triple ripples, phase distortion, over-correction, and one that reminds me of a pancake run through a frilling machine."

Channing looked at the collection of scrambled watches and shuddered. "Take'm away—*brrrrrr*."

Arden covered the uninspiring things with a tablecloth.

"Thanks," said Don.

"Do you think the judge'll forgive us?" asked Farrell.

"Don't say it," said Walt, bursting with laughter.

"I don't have to," chortled Don.

"They're both hysterical," explained Arden.

"Carbogen and Turkish bath," said Don. "And quick! Arden, call us a taxi."

"You're both taxis," giggled Arden. "O.K., fellows. Can do." She went to the phone and started to call.

Farrell looked uncomprehendingly at Walt and then at Don, and shook his head. "Mind telling me?" he pleaded.

"Wes, you're a million!" roared Channing, rolling on the floor.

Farrell turned to Arden.

"Let them alone," she said. "Something probably pleases them highly. We'll find out later—Yes? Operator? Will you call a cab for Room 719? Thanks."

Attorney Tinkin faced Judge Hamilton with a slight smile. "Prior to your ruling, I wish to present you with your watch. Also I ask permission to sum up my case—an act which I was unprepared to do last evening."

Hamilton reached for the watch, but Tinkin kept it.

"You may state your case—but it will make little difference in my ruling unless you can offer better evidence than your opponent."

"Thank you," said Tinkin. He made a show of winding the watch, and he set it accurately to the court clock on the wall. "Your honor, a telegram is a message. It requires energy for transmission. A letter also requires energy for carrying and delivery. A spacegram requires the expenditure of great energy to get the message across. The case in hand is this: If the energy is expended in maintaining the contact, then communications are involved. But when the energy is expected to be used on the other side—and the energies transmitted are far above and beyond those necessary for mere maintenance of contact, it then may be construed that not the contact but the transmittal of energy is desired, and power transmission is in force."

Tinkin swung Hamilton's watch by the chain.

"The matter of sending flowers by telegram is not a matter of taking a bouquet to the office and having the items sent by electricity to Northern Landing. A message is sent—an order to ship or deliver. It makes no difference whether the order be given in person or sent by spacegram. It is a communication that counts. In this device, a communication is sent which directs the device to produce a replica of the transmitted object. Ergo it must fall under the realm of communications. I will now demonstrate this effect, and also one other effect which is similar to telegraphic communications."

Tinkin ignored Hamilton's outstretched hand, and put the watch in the cabinet. Hamilton roared, but Tinkin held up a hand to stop him. "I assure you that this will cause no ill effects. We have repaired the damage."

"For every minute of delay between now and the moment I receive my watch, I shall fine you one hundred dollars for contempt of court."

"Well worth it," smiled Tinkin.

Channing pressed the switch.

Click! went the receiver, and from a slide, Channing removed the judge's watch. With a flourish, he started it, and handed it to the judge, who glared.

"Now," added Tinkin, "I wish to add—

Click!

"—two objects may be similar in form—

Click!

"—but can not be identities!

Click!

"However, two communications—

Click!

"—may be dissimilar in form—

Click!

"—but identical in meaning!

Click!

"We have before us—

Click!

"—a condition where—

Click!

"—identical messages are—

Click!

"—being reproduced in identical form—

Click!

"—just like a bunch of—

Click!

"—carbon copies!"

Click!

"The production rate of which—

Click!

"—will be high enough—

Click!

"—to lower the cost—

Click!

"—of this previously rare item—

Click!

"—until it is well within the reach of all."

Click!

"Just as in communications—

Click!

"—we may send an order—

Click!

"—directing the fabrication—

Click!

"—of several hundred similar items!

Click!

"And our supplier will bill us—

Click!

"—for them later!"

Brrr-rup!

"That last buzz or burp was a signal that we have reached the end of our matter bank. Our credit, for example, has run out. However, Dr. Channing is about to make a substantial deposit with the manufacturer, and we will resume operations later. I ask you—

Click!

"—can you do this with energy?"

Click!

"Stop that infernal—

Click!

"—machine before I have you all held for disrespect, perjury, contempt of court, and grand larceny!" yelled the judge.

Channing stopped the machine and started to hand out the carbon-copy watches to the audience, who received them with much glee. Kingman came to life at this point. He rose from his chair and started to object, but he was stopped by Tinkin who leaned over and whispered:

"My worthy and no doubt learned opponent, I'd advise that you keep your magnificent oratory buttoned tight in those flapping front teeth of yours. If we all get into that gadget—how would you like to fight ten or twelve of us?"

THE END.

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